# Envelope reprocessing of thermal radiation

Thomas Baxter

**Abstract**

The conditions for full or depth-dependent (partial) thermalisation of emission considering both scattering and absorption effects emerging from a hot central source through a reprocessing region typical of TDEs is derived.

**Improvements/continuations:**

- Relaxing the spherical symmetry requirements
- Adiabatic trapping in an outflowing wind
- Generalise beyond the conditions typical of TDEs: Inner engine temperatures $T_c \sim 10^5 K$, masses $M \sim M_\odot$, characteristic length scales $r_0 < 10^{15}$ cm.

## 1 Combined scattering and absorption

Consider the energy emitted by a volume dV over a solid angle $d\Omega$ a time dt:

$$dE = j_\nu dV d\Omega dt d\nu \tag{1}$$

where $dV = dAds$.

Considering the definition of the specific intensity $I_\nu$, we have the differential change in specific intensity along a given direction, $dI_\nu = j_\nu ds$. The absorption coefficient is defined by $dI_\nu = -\alpha_\nu I_\nu ds$.

The equation of radiative transfer for absorption and isotropic, coherent scattering[1]:

$$\frac{dI_\nu}{ds} = -\alpha_\nu^{ab} I_\nu + j_\nu - \alpha_\nu^s I_\nu + j_s \tag{2}$$

which can be rearranged using the definition of the emission coefficients

$$\frac{dI_\nu}{ds} = -(\alpha_\nu^{ab} + \alpha_\nu^s)(I_\nu - S_\nu) \tag{3}$$

where the source function with absorption and scattering:

$$S_\nu = \frac{j_\nu + \alpha_\nu^s J_\nu}{\alpha_\nu^{ab} + \alpha_\nu^s} \tag{4}$$

We define an opacity ratio $\epsilon_\nu$, which represents the probability that a random walk of a photon results in an absorption:

$$\epsilon_\nu = \frac{\alpha_\nu^{abs}}{\alpha_\nu^{es} + \alpha_\nu^{abs}} \tag{5}$$

[1]Coherent means that emission and absorption due to scattering is equal $\int \alpha_\nu^{es} I_\nu ds = j_s 4\pi$, where $j_s = \alpha_\nu^{es} J_\nu$

where $\alpha_\nu^{abs}$ and $\alpha_\nu^{es}$ are the frequency-dependent absorption (all processes) and electron scattering (isotropic) coefficients[2] respectively between frequencies $\nu + d\nu$.

It is convinient to define the optical depth $d\tau_\nu = \alpha_\nu ds$, we then have:

$$\tau_\nu = \int_{s_0}^{s} \alpha_\nu(s')ds \tag{6}$$

where $\tau(s_0) = 0$.

The effective optical depth to absorption is derived through random-walk arguements (Rybicki and Lightman, 1986) and given by $\tau_{\nu,tot}\sqrt{\epsilon_\nu}$[3] We then define the thermalisation radius $r_{\nu,th}$, which corresponds to the radius at which the effective optical depth for absorption is 1.

$$\tau_{\nu,th} = 1/\sqrt{\epsilon_\nu} \tag{7}$$

The photospheric radius $r_{ph}$ is defined as the surface of last scattering or most precisely last photon interaction. When the absorption opacity is large enough, the thermalisation radius aproaches the photospheric radius. If this is the case for the frequency bands of interest the emission can be modelled as a blackbody with an effective temperature $T_{eff}$ (assuming spherical symmetry):

$$L_{\rm bol} = 4\pi r_{\rm ph}^2 \sigma T_{\rm eff} 4 \tag{8}$$

For lower vales of $\epsilon_\nu$, the thermalisation radius can be significantly different across a frequency range, the emergent spectrum will not have a blackbody shape.

General form of $dL_\nu = j_\nu d\Omega dV$.

## 2 Example: free-free absorption in the context of TDEs

Consider a spherically symmetric, static, mass (gas) surrounding a hot central object. The density of the material declines from near the inner source, $r_0$, following a power-law profile $\rho \sim r^{-n}$. The surrounding gas is completely ionised, the scattering opacity, $\alpha_\nu^{es}$ , is dominated by electron scattering ($\alpha^{es} \gg \alpha^{ab}$) and is therefore frequency independent. Integrating from infinity to a radius $r$ gives the electron scattering optical depth:

$$\tau_{es} = (\frac{1}{n-1})\kappa_{es} r_0^n \rho_0 r^{1-n} \tag{9}$$

where $\alpha^{es} \propto r^{n-1}$.

The dominant source of absorption opacity in the NIR and at longer WL is free-free transitions. The free-electrons and ions are thermalised, therefore by kirchoff's law [4] the absorption emissivity ($j_\nu = \alpha_\nu^{abs} B_\nu$)

$$j_\nu^{ff} \propto T_e^{-1/2} Z^2 n_e n_i \exp(-h\nu/kT_e) \tag{10}$$

eq 5.18a Rybicki and Lightman 1986. In the regime $h\nu \ll k_B T$ gives the scaling $\alpha_\nu^{abs} \propto \rho^2 \nu^{-2}$.

The thermalisation depth at frequency $\nu$ is $\tau_{\nu,th} \approx \sqrt{\frac{\alpha^{es}}{\alpha_\nu^{abs}}} \propto \rho^{-1/2}\nu^1$. Equating to (9) the thermalisation radius scales as $r_{th} \propto \nu^{2/(2-3n)}$.

[2]The absorption coefficient can be defined as $\alpha_\nu = n\sigma_\nu$ [cm$^{-1}$] or equivalently $\alpha_\nu = \rho\kappa_\nu$.

[3]$\tau_{\nu,tot} = \int(\alpha_\nu^{abs} + \alpha_\nu^{es})ds$, is the total optical depth of absorption and scattering combined.

[4]Kirchhoff 's law for thermal radiation is that the source function of matter in thermal equilibrium is the Planck function

Now we can estimate the luminosity[5], neglecting constants and focusing just on the frequency dependence[6].

$$L_\nu \sim \int_{r_{\nu,th}}^{\infty} j_\nu^{ff} r^2 dr \tag{11}$$

Power spectrum then depends on frequency for a given power-law density profile.

$$\nu L_\nu \sim \nu^{\frac{8-7n}{2-3n}} \tag{12}$$

Slope of the blackbody spectrum given by rayleigh-jeans $\nu L_\nu \propto \nu^3$, therefore all density profiles $n > 2/3$ result in a shallower spectrum in the valid frequency band (NIR in TDEs).

## References

George B. Rybicki and Alan P. Lightman. *Radiative Processes in Astrophysics*. June 1986.

[5] $dL_\nu = j_\nu d\Omega dV$

[6] In reality the luminosity will carry a dependence on the electron temperature $T_e$